Problem-Solution Fit

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Small-to-medium scale precision farmers. • Regional agricultural consultants/agronomists. • Cooperative farm network operations managers.	**6. CUSTOMER LIMITATIONS [CL]** • **Budget:** Low capital for heavy upfront infrastructure updates. • **Devices:** Variable mobile network signals & mid-range smartphones. • **Tech literacy:** Prefers simple dashboards over raw code or scripts.	**5. AVAILABLE SOLUTIONS [AS]** • **Traditional Soil Lab Tests:** Highly accurate (+) but take days to deliver data and lack real-time visibility (-). • **Basic Meteorological Apps:** Free access (+) but lack hyper-local crop or soil context (-).
2. PROBLEMS / PAINS + FREQUENCY [PR] • Unpredictable crop damage from unrecognized soil nutrient drops. • **Frequency:** Occurs dynamically during intense changing growth cycles (weekly basis).	**9. PROBLEM ROOT / CAUSE [RC]** • Lack of continuous, localized telemetry mapping systems. • Delayed identification of moisture & macro-nutrient (NPK) baseline deviations before physical damage visible.	**7. BEHAVIOR + INTENSITY [BE]** • Manual spot-checking of massive fields; reactive application of generic fertilizers once discoloration shows. • **Intensity:** Critical; results in high yield losses.
3. TRIGGERS TO ACT [TR] • Sudden shifting localized weather forecasts. • Visible signs of leaf chlorosis or moisture stress in adjacent regional crop networks.	**10. YOUR SOLUTION [SL]** **OptiCrop Optimization Engine:** An integrated web platform connecting IoT tracking logs with real-time predictive ML alerts. Dispatches proactive NPK warnings within <20% variance to guarantee optimal yield control.	**8. CHANNELS OF BEHAVIOR [CH] - ONLINE** • Mobile WhatsApp/SMS telemetry alerts. • Progressive Web App (PWA) client views. • Agronomy forum messaging updates.

4. EMOTIONS BEFORE / AFTER [EM]

- **Before:** Anxious, uncertain, reactive to climate shifts.
- **After:** Empowered, confident, proactive through structural system metrics.

[OptiCrop Analytics Framework]

8. CHANNELS OF BEHAVIOR [CH] - OFFLINE

- Local agricultural cooperative meets.
- Word-of-mouth advisor field updates.
- Printable performance metrics sheets.